Student Academic Performance Analysis

Project ID: DA-003
Domain: Education Analytics

1. Objective

To analyze student marks, attendance, and study habits to identify patterns associated with academic performance and explore where additional academic support may be useful.

2. Tools and Technologies

- Python
- Google Colab / Jupyter Notebook
- Pandas
- Matplotlib
- Seaborn
- Synthetic student dataset

3. Methodology

1. Created a synthetic dataset containing 10 student records.
2. Organized Maths, Science, English, attendance, and study-hour data using Pandas.
3. Calculated mean, median, and pass percentage for each subject.
4. Created bar charts and box plots to compare subject performance.
5. Used scatter plots and Pearson correlation to examine associations between attendance, study hours, and average marks.
6. Checked for missing values, duplicate records, and invalid values.

4. Key Findings

- The overall average mark was approximately 68.03 out of 100.
- Science had the highest subject average at 69.4.
- Maths had an average of 66.4, while English averaged 68.3.
- Average attendance was 78.6%.
- Average reported study time was 2.8 hours per day.
- The attendance–marks correlation was approximately 0.99.
- The study-hours–marks correlation was approximately 0.95.

5. Conclusion

The synthetic dataset showed strong positive associations between attendance and average marks, and between study hours and average marks. Science had the highest average score, while Maths had the lowest average score in this sample.

These findings describe the generated dataset only. Correlation does not establish causation, and the small synthetic sample cannot be used to draw conclusions about an actual student population.

6. Recommendations

- Review attendance patterns alongside individual student performance.
- Offer additional subject support where students are struggling.
- Encourage consistent study routines and monitor whether students find them helpful.
- Use real, anonymized student data to validate these patterns before making institutional decisions.
- Evaluate academic support interventions using appropriate follow-up data.

7. Limitations

The dataset contains only 10 synthetic student records. It does not represent a real educational institution. Correlations may be unusually strong in a small, constructed dataset and should not be generalized to real students.

8. Future Scope

The project can be expanded using a larger anonymized dataset, additional subjects, assignment scores, and term-wise performance. Future analysis could examine whether observed patterns persist across different groups and time periods.